

Zerodha & Share Market — Basic Understanding

Beginner-friendly notes for understanding a Zerodha-style stock trading project

1. Share kya hota hai?

Maan lo ek company ABC Ltd. hai. Company apni ownership ko chhote parts mein divide karti hai, jinhe shares kehte hain. Agar total 10 lakh shares hain aur tumhare paas 1,000 shares hain, toh tum company ka ek bahut chhota ownership fraction hold karti ho.

2. Share ki price kyun change hoti hai?

Stock price exchange par buyers aur sellers ke orders ke interaction se discover hoti hai. Buying demand/selling pressure aur available liquidity ke according price move kar sakti hai.

3. Stock Exchange kya hota hai?

India mein major stock exchanges NSE (National Stock Exchange) aur BSE (Bombay Stock Exchange) hain. Retail user usually broker ke platform ke through order place karta hai.

4. Zerodha kya hai?

Zerodha ek stock broker/platform hai jiske through users stocks search, buy/sell orders place, orders track, holdings aur positions dekh sakte hain. Tumhara project iska simplified simulation hai.

5. Demat Account vs Trading Account

Demat account mein securities electronically hold hoti hain. Trading account ka use buy/sell transactions ke liye hota hai. Beginner mental model: Trading account = transactions; Demat account = securities held.

6. BUY ka basic workflow

User buy order place karta hai → broker order receive/route karta hai → exchange par matching hoti hai → agar order execute hota hai, order status update hota hai → delivery trade ke case mein applicable settlement ke baad holding reflect hoti hai.

7. Order kya hota hai?

Order broker ko diya gaya instruction hai, jaise BUY TCS, quantity 2. Order placed hona aur order executed hona alag concepts hain. Order pending, executed, rejected ya cancelled ho sakta hai.

8. Market Order vs Limit Order

Market order mein user execution ko priority deta hai aur exact price guarantee nahi hoti. Limit order mein user maximum buy price ya minimum sell price specify karta hai; suitable market condition aane par hi execution ho sakta hai.

9. Orders, Holdings aur Positions — sabse important distinction

Concept	Meaning	Example
Orders	User ne kya instruction diya	BUY TCS 10; SELL TCS 4
Holdings	User ke owned delivery shares	TCS → 10 shares

Positions	Current trading exposure / net position	BUY 10 – SELL 4 = +6
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10. Complete Buy → Sell example

Suppose TCS price is ₹3,000 and user buys 5 shares.

BUY TCS 5 → Order created → Order executes → Delivery holding eventually becomes TCS = 5.

Later user sells 2 shares:

Orders: BUY TCS 5 + SELL TCS 2

Remaining holding: TCS = 3

11. P&L; (Profit & Loss)

Example: Buy TCS at ₹3,000, current price ₹3,200, quantity 5. Approx unrealized P&L; = $(₹3,200 - ₹3,000) \times 5 = ₹1,000$. If current price is ₹2,800, P&L; = $-₹1,000$.

12. Average Price

If 5 shares are bought at ₹3,000 and another 5 at ₹3,200: total cost = ₹31,000; total quantity = 10; average price = ₹3,100.

13. Market Capitalization

Market cap is broadly calculated as outstanding shares $\times$ current market price. Example: 10 crore shares $\times$ ₹100 = ₹1,000 crore market capitalization.

14. IPO

IPO (Initial Public Offering) is when a company offers shares to the public for the first time. After listing, shares can trade in the secondary market through an exchange.

15. Tumhare Zerodha Clone ka architecture

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React UI
↓ Axios
Express API
↓
MongoDB
■■■ Orders collection
■■■ Holdings collection

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Optional: Positions can be calculated/maintained separately.

16. Tumhare project mein Buy/Sell ka simplified workflow

BUY:

User clicks Buy → Axios request → Express route → store BUY order → update UI.

SELL:

User clicks Sell from Holdings → create SELL order → update/remove holding in the simplified simulation → show SELL order in Orders.

17. Real Zerodha vs Tumhara Clone

Real-world flow:

User → Broker → Exchange → Order matching → Execution → Settlement → Holdings/Positions.

Your clone:

User → React → Axios → Express → MongoDB.

The clone does not actually connect to NSE/BSE or execute trades. It simulates the workflow for learning full-stack development.

18. Interview-ready mental model

Share = company ownership ka unit.

Exchange = buyers/sellers ke orders match hone ka marketplace.

Broker = user aur market infrastructure ke beech platform/intermediary.

Order = buy/sell instruction.

Execution = order actually fill hona.

Holding = owned delivery shares.

Position = current trading exposure/net position.

P&L; = profit/loss.

19. One-line project explanation

"I built a simplified stock trading platform inspired by Zerodha to understand full-stack development concepts such as React UI, REST APIs, MongoDB persistence, order management, holdings, and simulated buy/sell workflows."